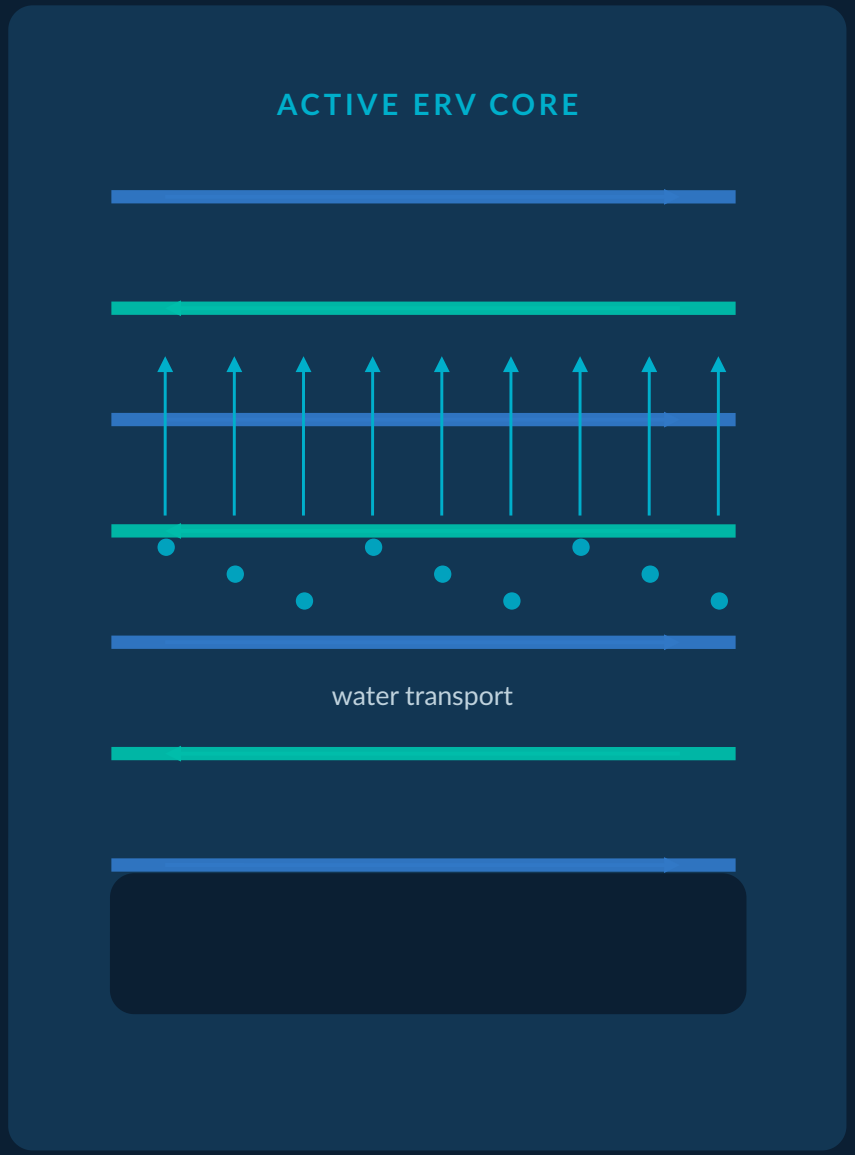
HEIQ / XMEMBRANE

Electro-Osmotic Water Transport Integrated into an ERV

Coupled model • O30 operating point • Validation path

MODEL SYNTHESIS | ENGLISH DECK | SEPTEMBER 2026

A physically based screening model that links moist-air thermodynamics, passive recovery, local membrane transport and cooling-coil power.



Prepared from the controlled model description and selected final result set

EXECUTIVE TAKEAWAY

The model identifies a promising—but conditional—energy window

Selected O30 operating point; active membrane integrated directly into the ERV core.

418.4
kW
Known modeled total power
Active ERV + EO + AC

125.34
kW
Conditional power window
vs. passive ERV + AC

5,899
m^2
Selected active membrane area
at 1.0 V terminal voltage

23.05
%
Reduction in known modeled power
EO-on vs. EO-off

WHAT THE RESULT MEANS

- The passive ERV delivers the largest established reduction: 622.66 kW versus AC-only.
- EO adds 1,228.3 kg/h of incremental water-transfer duty at O30.
- Reduced latent load can lower both condensate duty and compressor lift.

CLAIM BOUNDARY

This is not yet a validated net saving.

Receiver, thermal management, power electronics, manifolds and additional hardware must collectively remain below 125.34 kW.

Decision framing: validate the membrane vector and close the omitted subsystem losses before making a net-savings claim.

SYSTEM COMPARISON

Three systems isolate what passive recovery and EO each contribute

1 • AC only

1,166.400 kW

Outdoor air is treated directly by the cooling coil.

BASELINE

2 • Passive ERV + AC

543.741 kW

Sensible and latent recovery reduce the coil duty before cooling.

EO OFF

3 • Active ERV + EO + AC

418.400 kW

Voltage-driven water transport is added inside the same ERV membrane.

EO ON

622.659 kW Passive ERV reduction vs. AC only

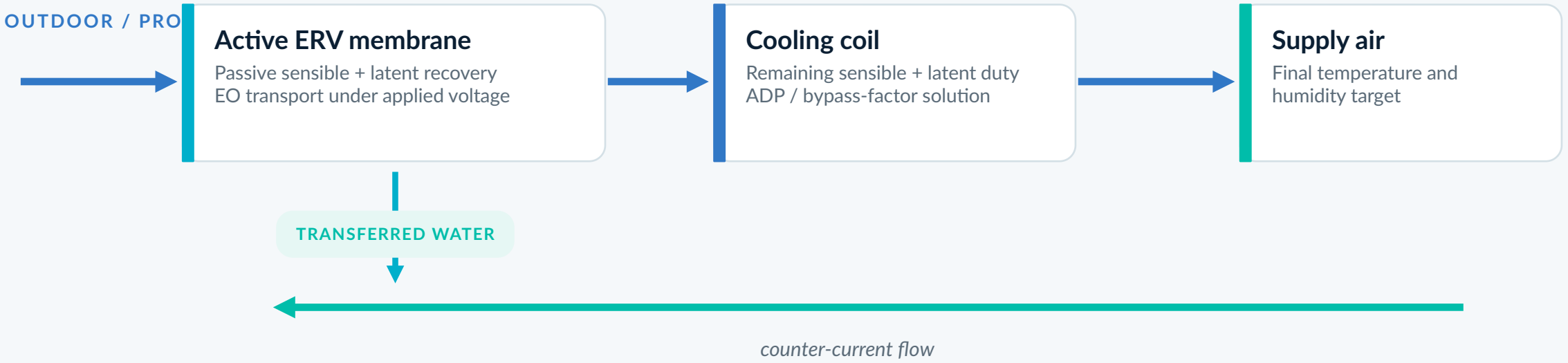
125.340 kW Incremental EO window vs. EO off

Do not label the full AC-only → active-ERV difference as “EO savings.”

SELECTED ARCHITECTURE

EO is embedded in the ERV membrane—not added as a separate downstream stage

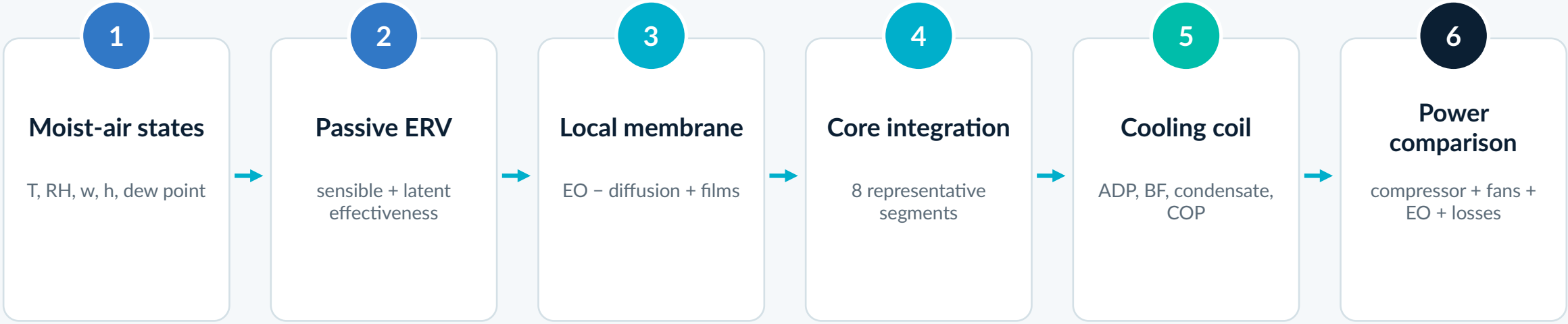
The cooling coil remains in the system and completes cooling and dehumidification after O30.



Physical mechanisms

HEAT	warm → cool	DIFFUSION	activity gradient	EO	ionic current drags water

Local membrane physics is evaluated inside a complete HVAC energy balance



Core modeling principle

Every kilogram of water removed from the process air must appear in the exhaust stream, and every included electrical load is carried into the final comparison.

Forward model: measured membrane vector → system outcome

Inverse design: required system outcome → target material region

LAYER 1

Psychrometrics and passive ERV recovery set the boundary conditions

Moist-air state vector

Temperature T
Relative humidity RH
Humidity ratio w
Enthalpy h, dew point and specific volume

$$w = m_water \,/\, m_dry\ air$$

$$h \approx 1.006T + w(2501 + 1.86T)$$

$$\dot{m}w = \dot{m}da\ (win - wout)$$

Passive ERV effectiveness

Sensible effectiveness εs updates temperature.
Latent effectiveness εl updates humidity ratio.

$$Tout = TOA - \varepsilon s(TOA - Texh)$$

$$wout = wOA - \varepsilon l(wOA - wexh)$$

$$Pfan = V\ \Delta p \,/\, \eta fan$$

BALANCE CHECKS

- ✓ Process / exhaust water balance
- ✓ Enthalpy balance
- ✓ No untreated condensation state
- ✓ Fan pressure-drop penalty

Output: local T and water-activity boundaries for the active membrane.

LAYER 2

Net water transport is EO drag minus opposing diffusion

$$\mathrm{Nw} = \mathrm{tw} \cdot \mathrm{j} \,/\, \mathrm{F} \;-\; \mathrm{Dw(aw)}\ \mathrm{Sw(aw)} \cdot \mathrm{daw/dx}$$

ELECTRO-OSMOTIC TERM

$$\mathrm{NEO} = \mathrm{tw} \cdot \mathrm{j} \,/\, \mathrm{F}$$

Ionic current carries additional water through the membrane.

DIFFUSIVE TERM

$$\mathrm{Ndiff} = -\mathrm{Dw}\ \mathrm{Sw} \cdot \mathrm{daw/dx}$$

Activity gradients can oppose the desired EO direction.

tw is an input in the current inverse-design model—not a predicted material property.

High tw values remain hypothetical until measured in vapor–membrane–vapor conditions.

Practical gross EO flux: $\mathrm{JEO,gross} \approx 0.672 \cdot \mathrm{tw} \cdot \mathrm{j}$ $[\mathrm{g\ m^{-2}\ h^{-1}}]$

LAYER 3

Voltage control couples transport to hydration-dependent resistance

$$dV/dx = j / \sigma(aw) + tw \cdot RT/F \cdot d\ \ln(aw)/dx$$

Ohmic drop

Vohm = j · ASRmembrane

Thickness ↑ or conductivity ↓ → voltage ↑

Water polarization

Vwater = tw · RT/F · ln(ar/as)

Higher tw also raises reversible water work

Series + fixed losses

j · ASRseries + Vfixed

Electrodes, contacts and electronics matter

At fixed terminal voltage, current density is solved—not freely selected.

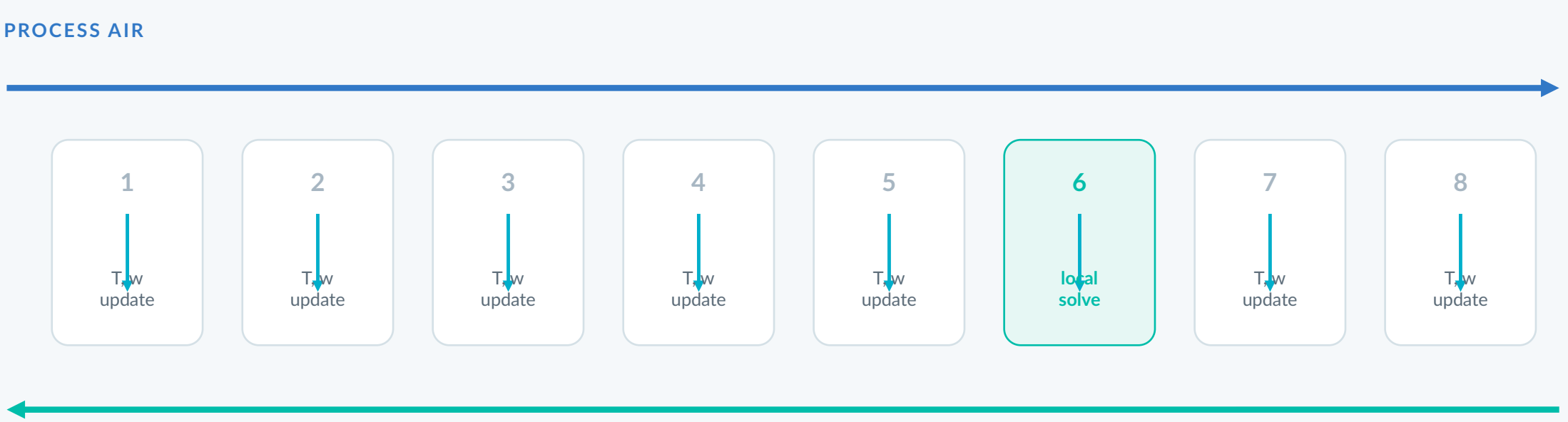
MATERIAL VECTOR

$$\{ tw, D, S, \sigma, L, ASR \}$$

$D(aw)$	water diffusivity
$\sigma(aw)$	ionic conductivity
$S(aw)$	differential sorption capacity

Drying can reduce σ, increase resistance, lower current and raise the required voltage.

Gas films and changing air states make local solutions essential



$$Nw = Ks(as,bulk - as,surface)$$

$$Ai = \dot{n}w,i \;/\; Nw,i$$

$$P\ EO = \Sigma\ Vi\ ji\ Ai$$

$$Nw = Kr(ar,surface - ar,bulk)$$

Illustrative boundary case: 16.80 mg/h gross Faraday rate → 12.37 mg/h net (73.6%) after films and back-diffusion.

Pre-drying can reduce condensate duty and improve compressor COP

Active ERV outlet
Lower w before coil

→

Coil ADP solution
BF mixes inlet and ADP states

→

Cooling load
ṁda(hin − hoff-coil)

→

Variable COP
scaled Carnot relation

→

Electrical power
Qcool / COP + reheat

COIL EQUATIONS

$$\mathrm{woff} = \mathrm{BF} \cdot \mathrm{win} + (1 - \mathrm{BF}) \cdot \mathrm{wADP}$$

$$\dot{m}\mathrm{cond} = \dot{m}\mathrm{da}(\mathrm{win} - \mathrm{woff})$$

DOUBLE BENEFIT

- ✓ **Less water must be condensed**
- ✓ **Higher evaporating temperature may increase COP**

The coil remains necessary at O30.

SELECTED OPERATING POINT

O30 is an intermediate humidity state—not the final supply condition

Passive ERV outlet

13.215

g/kg dry air

ACTIVE ERV →

O30 active ERV outlet

8.886

g/kg dry air

COOLING COIL →

Final target: 20°C / 50% RH*

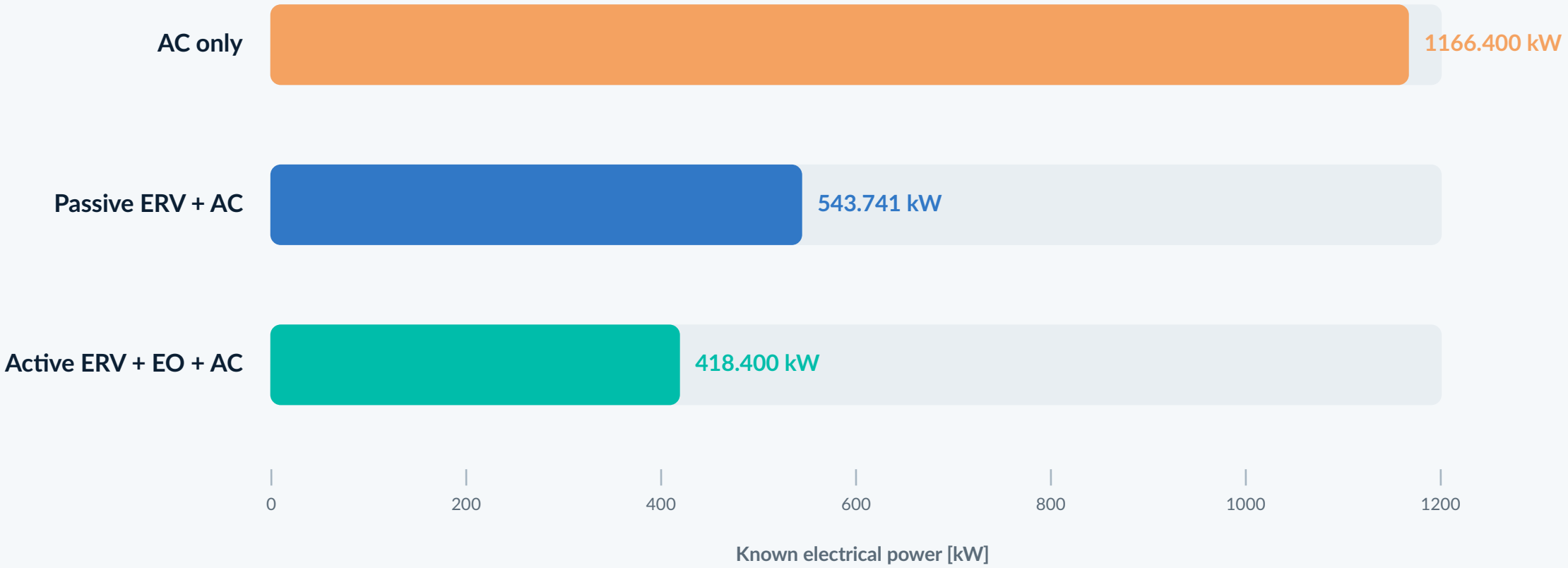
≈ 7.3

g/kg dry air

Interpretation EO removes most of the remaining moisture after passive recovery; the coil still cools and dehumidifies to the final supply state.

**Approximate humidity ratio at 101.3 kPa; the model uses the exact psychrometric state.*

Known power falls by 125.34 kW versus the same ERV with EO switched off



Passive recovery: −622.659 kW

Incremental EO window: −125.340 kW | −23.05%

Separate total core transport from the incremental EO duty

Water transport through active ERV core



Total = 4,365.9 kg/h

740
g m⁻² h⁻¹
Average total core flux

208
g m⁻² h⁻¹
EO increment / area

200–250
g m⁻² h⁻¹
Coupon-test target range

KNOWN ACTIVE-SYSTEM SPLIT

316.910 kW remaining AC power

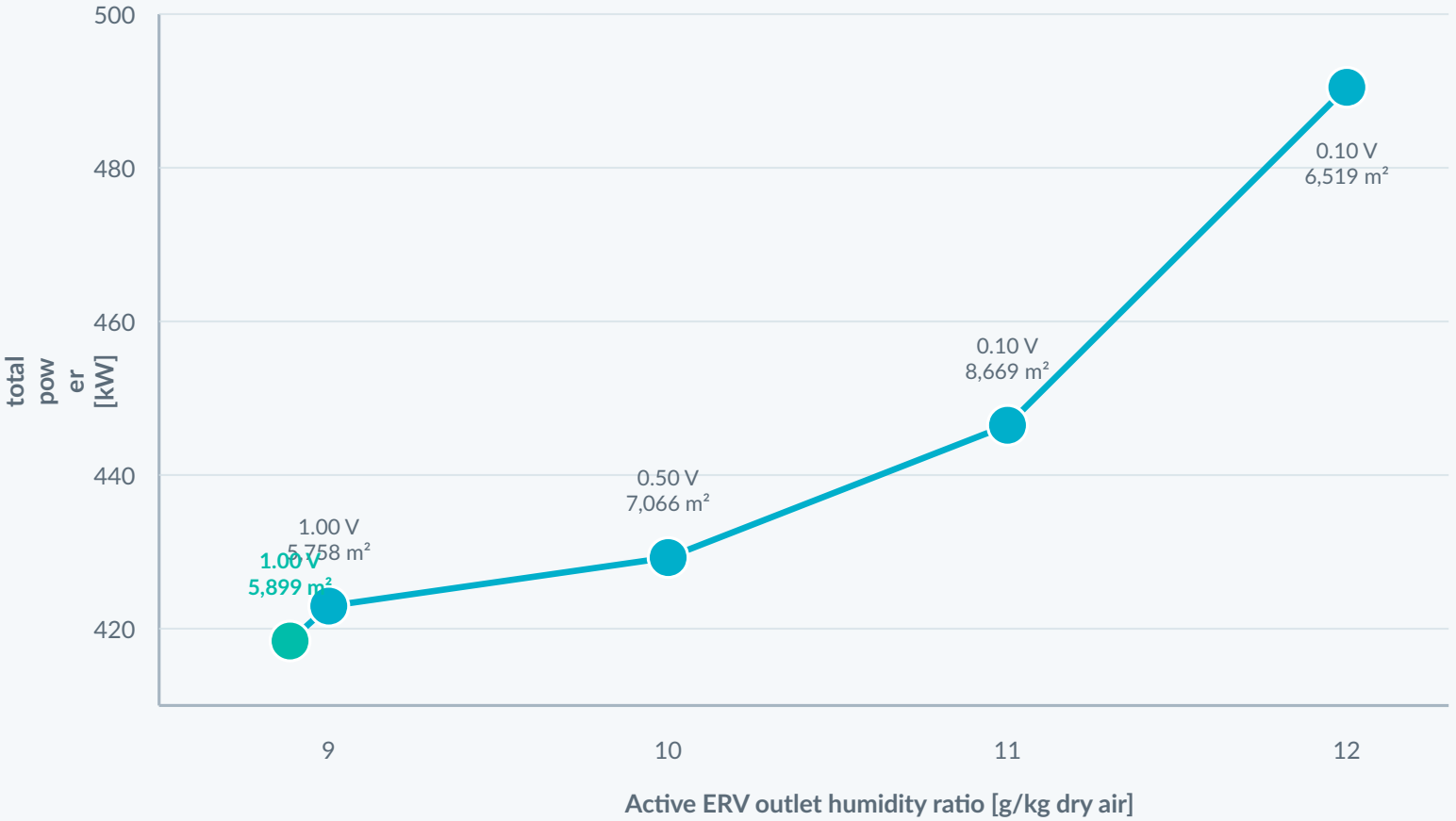
≈101.490 kW included active-core and other contributions

418.400 kW
known modeled total

Unmodeled losses must fit within 125.34 kW.

SELECTED RESULT FRONTIER

Deeper drying trades voltage, membrane area and system power



O30 SELECTION

8.886 g/kg

1.00 V • 5,899 m² • 418.400 kW

The optimum is not simply the lowest voltage or smallest area. Higher voltage can reduce required area while increasing electrical stress and power.

Screening area cap: 9,000 m²

Full dehumidification required at least ≈17,253 m² in the valid screening—outside the selected area cap.

A high-value screening model—not a validated commercial module

WHAT THE MODEL IS

- ✓ Physically based research model
- ✓ 1D transport through membrane thickness
- ✓ Segmented counter-current ERV core
- ✓ Coupled water balance between air streams
- ✓ Voltage-controlled ionic transport
- ✓ Reduced but realistic coil / energy model
- ✓ Forward + inverse-design screening tool

WHAT IT IS NOT

- ! Full 2D / 3D CFD or conjugate PDE model
- ! Validated commercial ERV module
- ! Validated electrochemical electrode model
- ! Dynamic aging, fouling or durability model
- ! Proof that high tw is achievable in vapor
- ! Closed receiver / drainage / heat-rejection design
- ! CAPEX or lifecycle-cost model

Use the model to define the experiment and the material target—not to skip them.

VALIDATION PATH

The next experiment must close transport, electrical and balance claims together

1. **Polarity-dependent net water flux**
   Demonstrate reversible directionality and quantify net flux.

2. **Effective tw in vapor**
   Derive from EO flux and current under representative RH.

3. **Conductivity / ASR at the same RH**
   Measure the electrical state that actually supports the flux.

4. **Passive back-diffusion**
   Run matched EO-off controls across the same activity gradient.

5. **Donor / receiver closure**
   Mass-balance both air streams; identify saturation or film limits.

6. **Stable operation at 1 V**
   Track current, heat, electrodes and performance over time.

Success criterion: measured membrane + hardware performance preserves a positive margin inside the 125.34 kW conditional window.

CONCLUSION

The model defines a credible target. The experiment must decide whether it is real.

SYSTEM	418.4 kW known modeled total at O30
MARGIN	125.34 kW conditional window vs. passive ERV + AC
HARDWARE	5,899 m^2 active membrane at 1 V
MATERIAL	Inverse-design vector still requires vapor validation

RECOMMENDED NEXT MOVE

Build the vapor coupon test around the incremental target:

200–250

g m^{-2} h^{-1} net, polarity-dependent water transport

paired with current, ASR, thermal and donor/receiver balance measurements.

APPENDIX A

Core equation map

PSYCHROMETRICS

$$h = 1.006T + w(2501 + 1.86T)$$
$$\dot{m}w = \dot{m}da(win - wout)$$

PASSIVE ERV

$$Tout = TOA - \varepsilon s(TOA - Texh)$$
$$wout = wOA - \varepsilon l(wOA - wexh)$$

MEMBRANE FLUX

$$Nw = tw\ j/F - Dw(aw)Sw(aw)\ daw/dx$$

VOLTAGE

$$Vterminal = Vohm + Vwater + jASRseries + Vfixed$$

CORE INTEGRATION

$$Atotal = \Sigma\ Ai$$
$$PEO = \Sigma\ Vi\ ji\ Ai$$

COOLING COIL

$$Qcool = \dot{m}da(hin - hoff)$$
$$Pcompressor = Qcool\ /\ COP$$

All balances are evaluated on a dry-air basis unless stated otherwise.

APPENDIX B

Gross EO flux scales linearly with tw and current density

At j = 10 A/m², before back-diffusion and gas-film losses

tw	Gross EO flux [g m^{-2} h^{-1}]
10	67
20	134
30	202
50	336
75	504
100	672

INTERPRETATION

$$J_{EO,gross} \approx 0.672 \cdot tw \cdot j$$

- Gross Faradaic transport is not the useful system flux.
- Back-diffusion and gas-film resistance reduce net transport.
- High tw also increases the water-polarization voltage term.
- The complete property vector—not a single number—sets feasibility.

Treat high tw values as design coordinates until measured.

APPENDIX C

Questions the audience is likely to ask

Is tw measured for the selected membrane?

No. It is an inverse-design parameter until measured under relevant vapor conditions.

Why not use liquid-cell values directly?

Hydration, morphology, conductivity and interfacial transport can differ radically between liquid and two gas streams.

Why is not all gross EO flux useful?

Back-diffusion, activity gradients and gas films oppose or limit the desired transport.

Why is the membrane area so large?

The air flow is ~150,000 CFM while local net flux is limited; compact ERVs use many thin parallel channels to create area.

Is 23.05% a documented saving?

No. It is a reduction in the known modeled sum; all omitted losses must fit inside 125.34 kW.

What would validate the concept?

Matched polarity tests, vapor tw, RH-resolved ASR, back-diffusion, mass balance, heat and stable 1 V operation.

Controlled references: docs/MODEL_CHAIN.md and the Final Result and Decision Report (v9, 31 Aug 2026).